AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 3 · STUDENT LESSON

Typed contracts and schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Architecture and data flow. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of typed contracts and schemas. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of typed contracts and **schema**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of typed contracts and **schema**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Architecture and data flow. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain typed contracts and **schema**s using the deciding step in this fictional service.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Typed contracts and schemas case. Separate observations, given facts and assumptions before choosing an explanation.

Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Typed contracts and schemas. Connect each labelled input, deciding step and result.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

Typed contracts and schemas — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python.	Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before processing.	Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Typed contracts and schemas. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Change A count to -1.

Predicted result

Prediction: Reject A on the non-negative value rule.

Why the result follows

Reason: The type is correct but the value is outside its accepted range.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Typed contracts and schemas to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Schema	A contract for structure and types
Runtime validation	Checking actual incoming values
Closed object	An object rejecting unexpected fields

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Typed contracts and schemas, predict a step, then compare with its explanation.

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python.

Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Worked explanation · Typed contracts and schemas

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Typed contracts and schemas

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def validate_request(value):
    if type(value) is not dict or set(value) != {'room_id', 'requested_count'}:
        raise ValueError('request fields')
    if type(value['room_id']) is not str or not value['room_id']:
        raise ValueError('room ID')
    if not nonnegative_integer(value['requested_count']):
        raise ValueError('requested count')
    return dict(value)

print(validate_request({"room_id": "A", "requested_count": 0}))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Typed contracts and schemas

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

The facts we will use

Supplied facts

A={sample_id:"jar-8",count:4}; B={sample_id:"jar-9",count:"4"}. No type conversion is

allowed.

AI ACADEMY / AI-6/3

Typed contracts and schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect A

The fields and integer count satisfy the contract.



2. Inspect B

The quoted count is a string, so B fails the type check.



3. Return status

Accept A; reject B before running the lab operation.

CHANGE ONE THING

Change A count to -1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect A	The fields and integer count satisfy the contract.
2. Inspect B	The quoted count is a string, so B fails the type check.
3. Return status	Accept A; reject B before running the lab operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **schema** checks declared structure. Looking like a number is different from having the required numeric type.

What this example does not establish

Keep the stated limit

Valid structure does not establish that the reported count is accurate.

Keep the stated limit

Read the labelled picture: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

Keep the stated limit

Change one thing: Change A count to -1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Typed contracts and schemas claim and explain a repair using the supplied evidence.

Schema-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Typed contracts and schemas with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Typed contracts and **schema**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Typed contracts and **schema**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Typed contracts and schemas. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def validate_request(value):
    if type(value) is not dict or set(value) != {'room_id', 'requested_count'}:
        raise ValueError('request fields')
    if type(value['room_id']) is not str or not value['room_id']:
        raise ValueError('room ID')
```

```
    if not nonnegative_integer(value['requested_count']):
        raise ValueError('requested count')
    return dict(value)

print(validate_request({"room_id": "A", "requested_count": 0}))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Typed contracts and schemas. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled typed contracts and **schema**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of typed contracts and **schema**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My limited claim and a useful next check**
>
> ___
>
> ___
>
> ___
>
> ___

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Typed contracts and schemas case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Typed contracts and schemas — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python.	Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before processing.	Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Classify counts 0, -1, "3", False under the stated contract.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Classify counts 0, -1, "3", False under the stated contract.

> **My independent procedure, result and limitation**
>
> _____
>
> _____
>
> _____
>
> _____

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Typed contracts and schemas, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

By the end, I can	How I will show it
Explain the mechanism and terminology of typed contracts and schemas	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Typed contracts and schemas without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of typed contracts and **schema**s

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept A; reject B before running the lab operation.

1. Inspect A

2. Inspect B

3. Return status

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return status Accept A; reject B before running the lab operation. A **schema** checks declared structure.

Looking like a number is different from having the required numeric type.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change A count to -1. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Reject A on the non-negative value rule.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject A on the non-negative value rule. The type is correct but the value is outside its accepted range.

Repair a missing building block

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

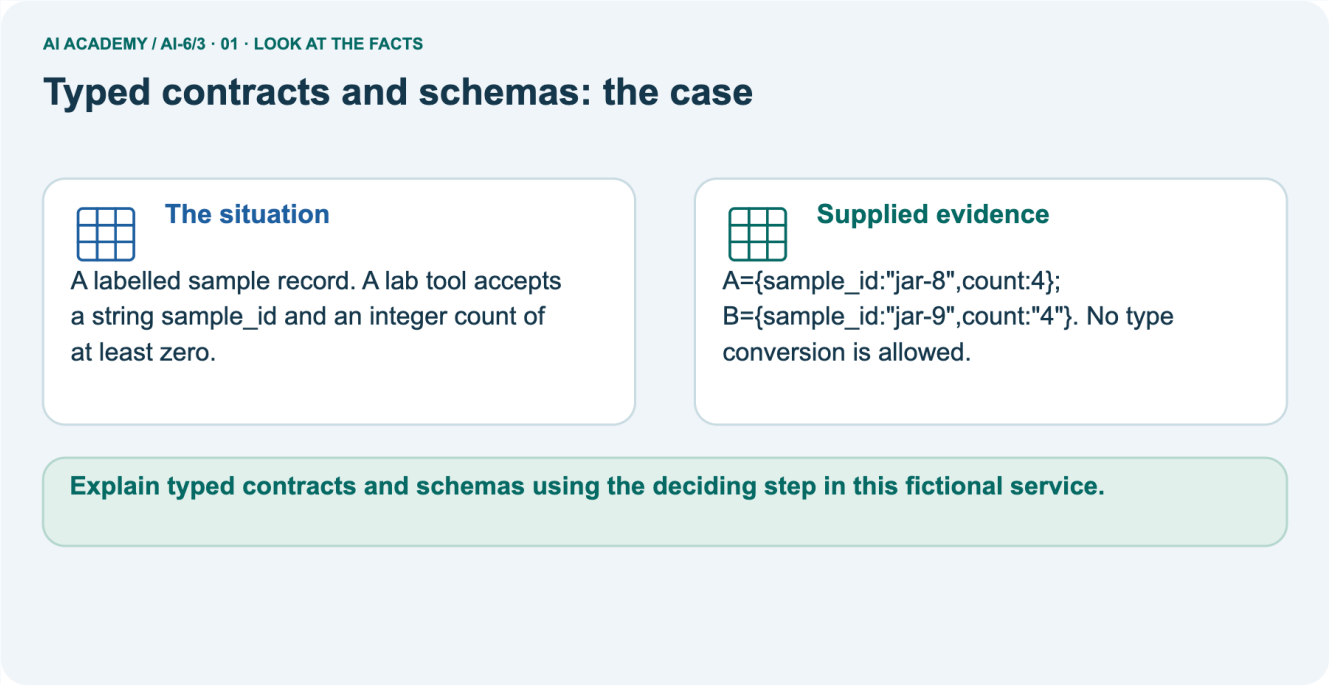
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Typed contracts and schemas graphic using its labels, deciding evidence and relevant limitation.

01. Typed contracts and schemas: the case

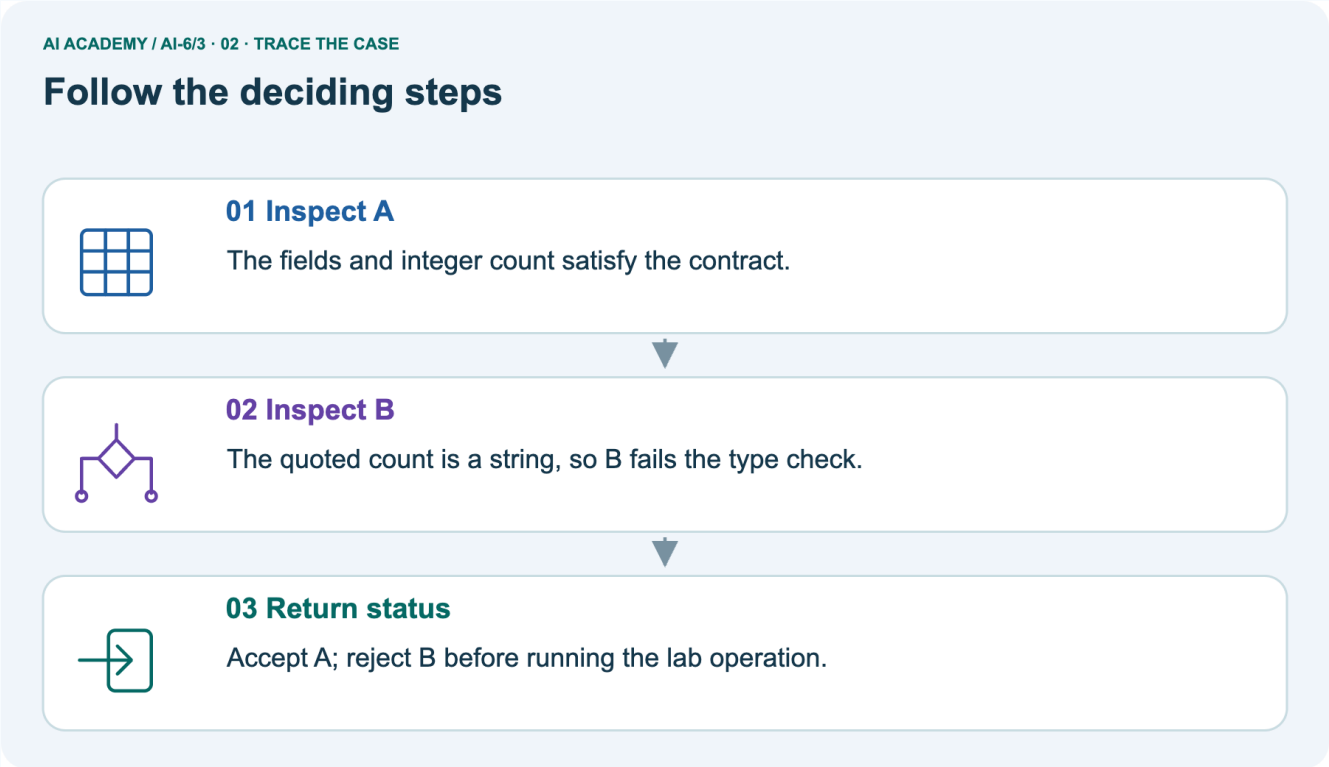


Follow the labelled picture with the matching explanation. Open the vector diagram

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero. Supplied facts: A={sample_id:"jar-8",count:4}; B={sample_id:"jar-9",count:"4"}. No type conversion is allowed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

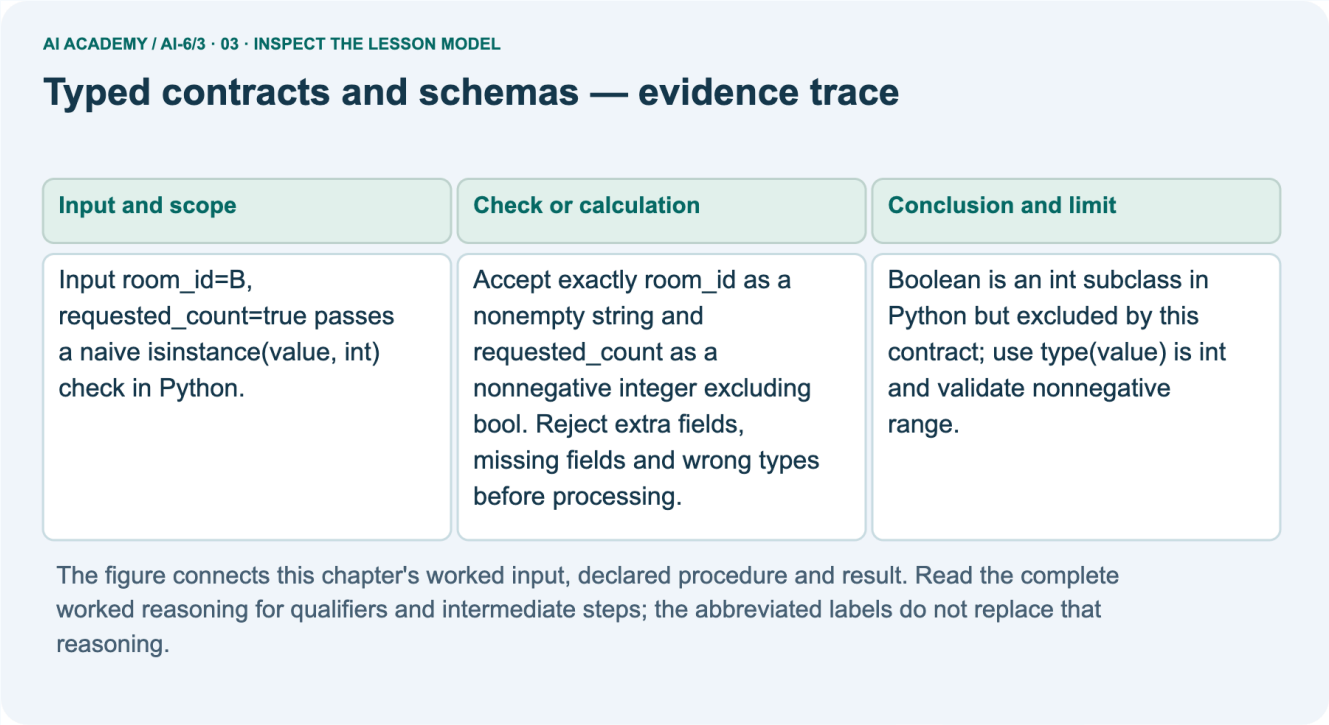


Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect A: The fields and integer count satisfy the contract. → Inspect B: The quoted count is a string, so B fails the type check. → Return status: Accept A; reject B before running the lab operation.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Typed contracts and schemas — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

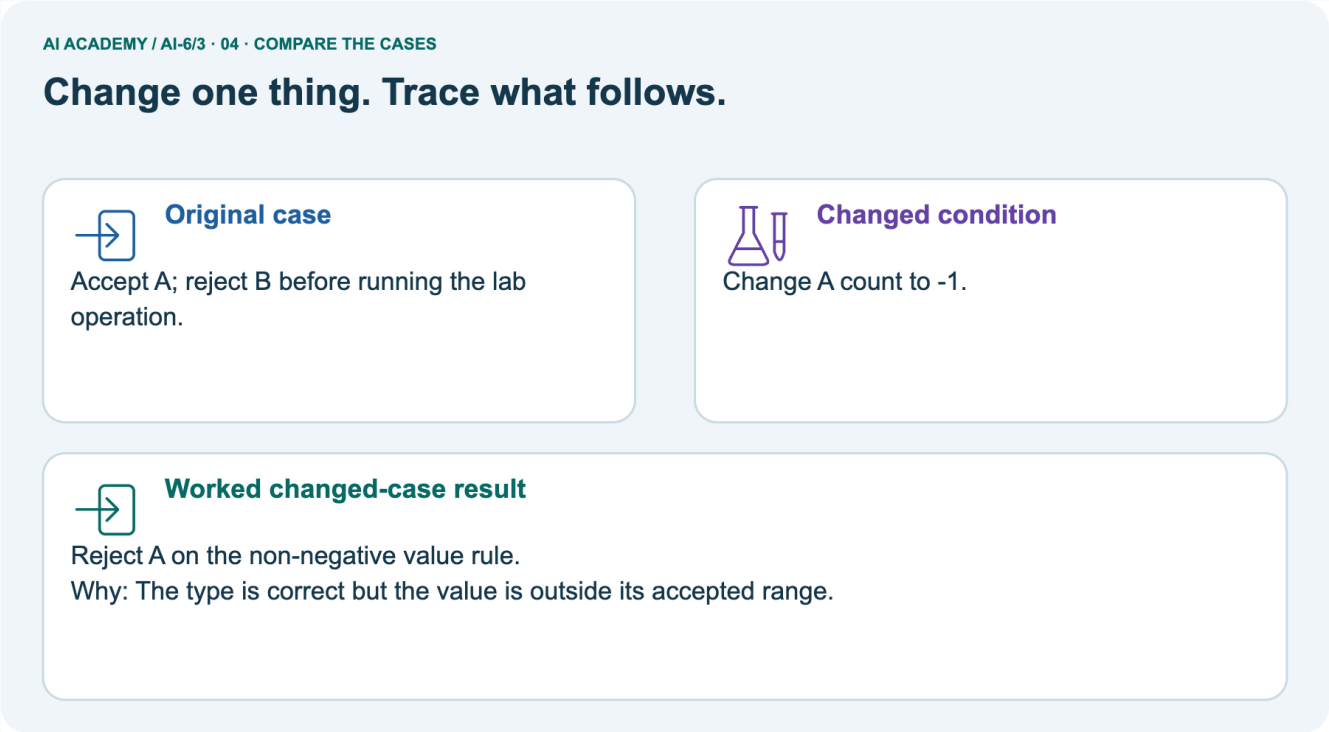
Typed contracts and **schema**s — **evidence** trace.

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. | Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool.

Reject extra fields, missing fields and wrong types before **process**ing. | Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



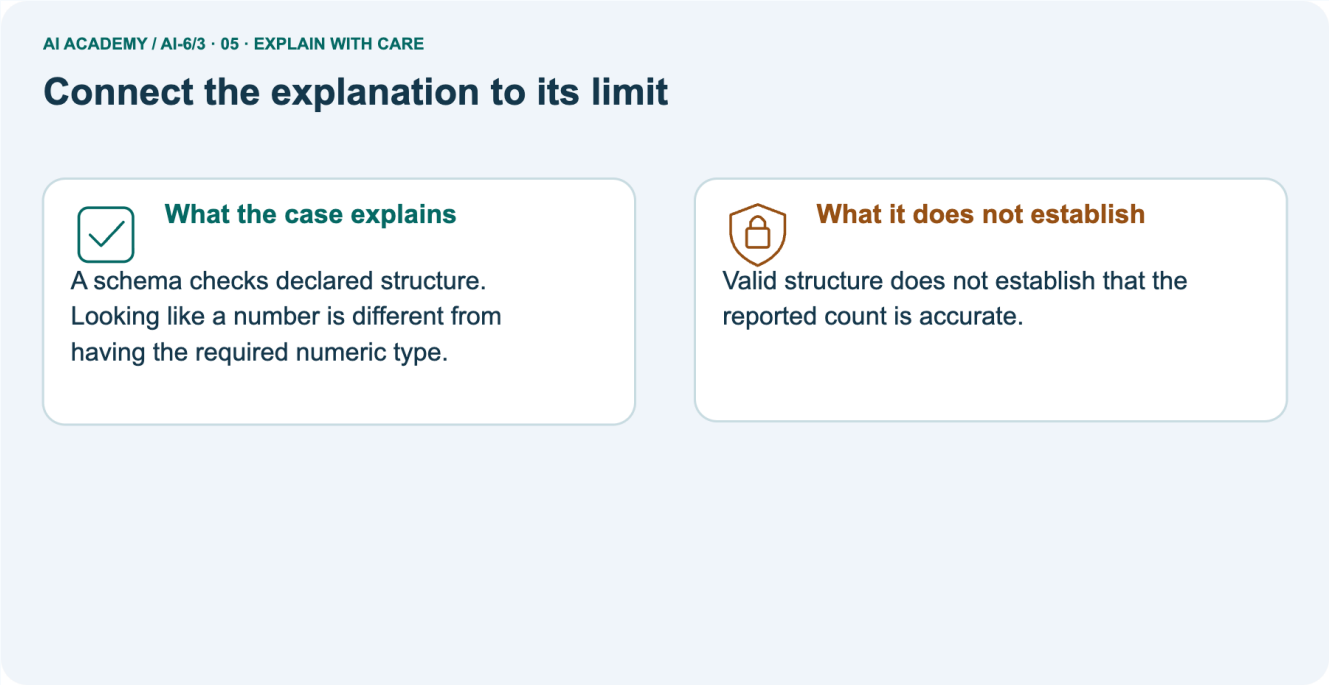
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept A; reject B before running the lab operation. Changed condition: Change A count to -1. Expected result: Reject A on the non-negative value rule.

Reason: The type is correct but the value is outside its accepted range.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

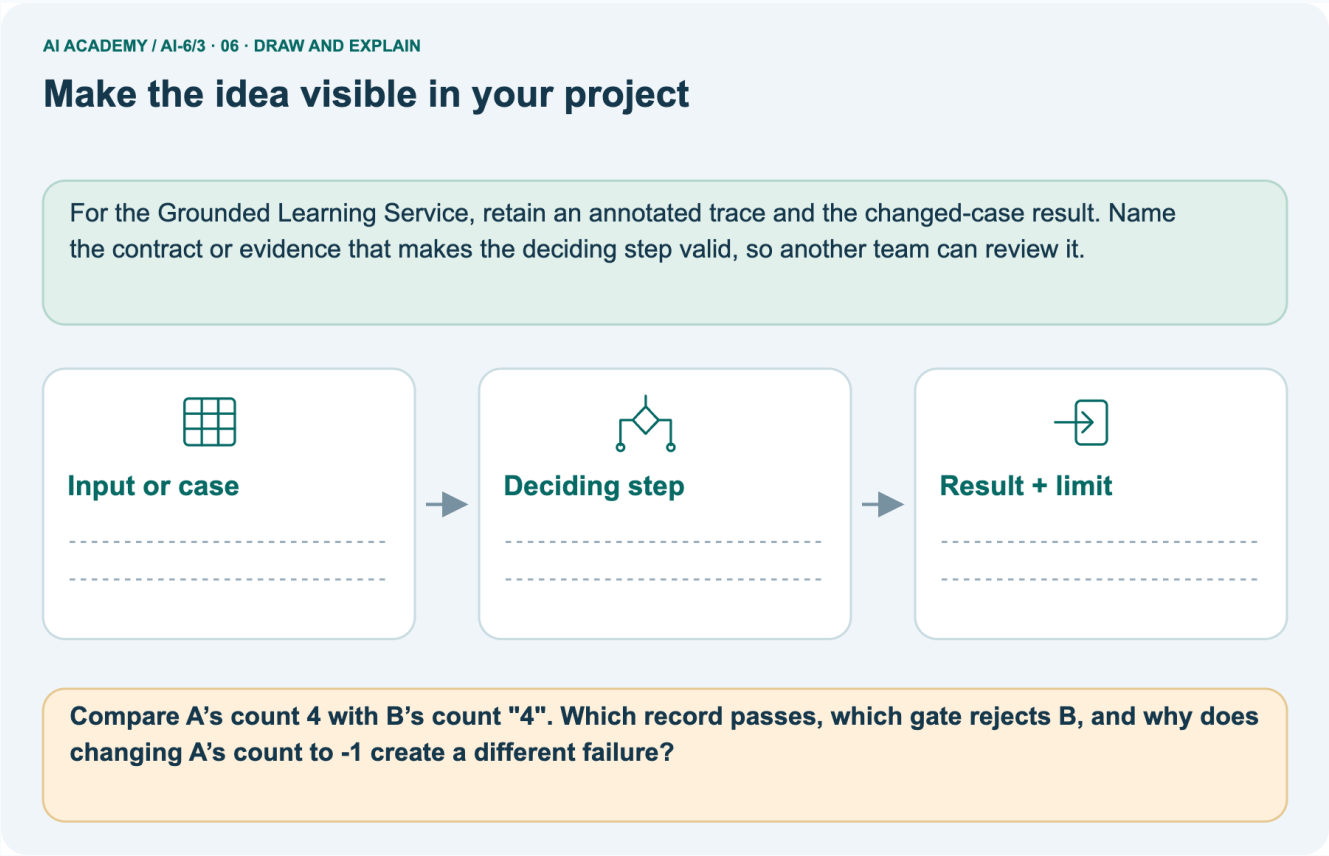


Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** checks declared structure. Looking like a number is different from having the required numeric type. Limit: Valid structure does not establish that the reported count is accurate.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.